Light

GCSE Computer Science – Data Representation: Numbers (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Data Representation Numbers

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Denary (decimal) is which number base?

- **A.** Base 10
- **B.** Base 8
- **C.** Base 2
- **D.** Base 16

Q2 (1 mark)

In hexadecimal, what letter represents the denary value 15?

- **A.** A

- **B.** D
- **C.** E
- **D.** F

Q3 (1 mark)

A nibble consists of how many bits?

- **A.** 2
- **B.** 4
- **C.** 8
- **D.** 16

Q4 (1 mark)

What is the binary representation of denary 7?

- **A.** 111
- **B.** 110
- **C.** 101
- **D.** 1000

Q5 (1 mark)

How many megabytes (MB) are in 1 gigabyte (GB)?

- A. 100
- B. 500
- C. 10,000
- D. 1,000

Q6 (1 mark)

What types of data can binary represent in a computer?

- A. Only numbers
- B. Only text
- C. Only images and sounds
- D. All data including numbers, text, images, and sounds

Q7 (1 mark)

Memory addresses in computers are typically displayed in which format?

- A. Binary
- B. Denary
- C. Hexadecimal
- D. Roman numerals

Q8 (1 mark)

What is the denary value of hexadecimal 1A?

- ○ **A.** 26
- ○ **B.** 20
- ○ **C.** 16
- ○ **D.** 30

Q9 (1 mark)

What is the maximum denary value that can be stored in 8 bits?

- ○ **A.** 128
- ○ **B.** 200
- ○ **C.** 256
- ○ **D.** 255

Q10 (1 mark)

Which unit of storage is the largest?

- ○ **A.** Megabyte
- ○ **B.** Kilobyte
- ○ **C.** Gigabyte

- D. Terabyte

Q11 (1 mark)

In the binary number 10000000, what is the denary value?

- A. 128
- B. 100
- C. 64
- D. 256

Q12 (1 mark)

How many different values can one byte represent?

- A. 8
- B. 100
- C. 512
- D. 256

Q13 (2 marks)

Calculate: 1011 + 0101 in binary.

- A. 1110

- B. 1111
- C. 10000
- D. 10001

Q14 (2 marks)

Convert the hexadecimal number 5D to binary.

- A. 0101 1101
- B. 0101 1100
- C. 0100 1101
- D. 0110 1101

Q15 (2 marks)

What is the result of left shifting the binary number 00010100 by 1 place?

- A. 00001010
- B. 00010100
- C. 01010000
- D. 00101000

Q16 (2 marks)

Convert denary 99 to binary.

- A. 0101 1111
- B. 0110 0001
- C. 0110 0011
- D. 0110 0100

Q17 (2 marks)

A video file is 750 MB. How many of these files can fit on a 6 GB USB drive?

- A. 4
- B. 6
- C. 8
- D. 10

Q18 (2 marks)

What is the result of right shifting 10100000 by 2 places?

- A. 00100100
- B. 00101000
- C. 01010000
- D. 10000010

Q19 (2 marks)

Convert denary 200 to hexadecimal.

- ○ **A.** B8
- ○ **B.** C8
- ○ **C.** D0
- ○ **D.** E0

Q20 (2 marks)

Calculate: 0001 + 0010 + 0011 in binary.

- ○ **A.** 0101
- ○ **B.** 0111
- ○ **C.** 0110
- ○ **D.** 1000

Q21 (2 marks)

Convert binary 1110 0010 to hexadecimal.

- ○ **A.** D2
- ○ **B.** E2
- ○ **C.** E4

- **D.** F2

Q22 (2 marks)

How many bytes are in 5 MB?

- **A.** 5,000
- **B.** 50,000
- **C.** 500,000
- **D.** 5,000,000

Q23 (2 marks)

To divide a number by 4 using binary shifts, how many right shifts are needed?

- **A.** 1
- **B.** 2
- **C.** 3
- **D.** 4

Q24 (3 marks)

Calculate the result of adding these three 8-bit binary numbers: 0100 0001 + 0001 1000 + 0010 0110

- **A.** 0111 1111

- B. 0111 0101
- C. 1000 0000
- D. 0110 1111

Q25 (3 marks)

A MAC address is displayed as 4A:3B:2C:1D:00:FF. Analyse what the last pair (FF) represents in denary and binary.

- A. FF = 240 denary = 1111 0000 binary
- B. FF = 250 denary = 1111 1010 binary
- C. FF = 255 denary = 1111 1111 binary
- D. FF = 256 denary = 1 0000 0000 binary

Q26 (3 marks)

A programmer uses right shifts to divide 50 by 8. Analyse the process and explain why the result differs from standard division.

- A. Right shift by 3 gives 6 (binary 00000110); this differs from 6.25 because binary shifts perform integer division, discarding any fractional part
- B. Right shift by 3 gives 7 (binary 00000111); shifts always round up
- C. Right shift by 8 gives 6; the shift amount equals the divisor
- D. Right shifts cannot be used for division

Q27 (3 marks)

An image is 4000 x 3000 pixels with 24-bit colour depth. Calculate the uncompressed file size.

- ○ **A.** 12 MB
- ○ **B.** 36 MB
- ○ **C.** 144 MB
- ○ **D.** 288 MB

Q28 (3 marks)

Calculate: 0100 0100 + 0010 0010 + 0001 0001 in binary.

- ○ **A.** 0110 0110
- ○ **B.** 0110 0111
- ○ **C.** 0111 0111
- ○ **D.** 1000 0000

Q29 (3 marks)

A student claims that hexadecimal 3G is a valid value. Analyse this claim and explain what the correct maximum two-digit hex value is.

- ○ **A.** 3G is valid; hexadecimal uses letters A through G
- ○ **B.** 3G is invalid because G is not a hexadecimal digit; the maximum two-digit hex value is FF (255 in denary)

- ○ **C.** 3G is invalid because hexadecimal only uses numbers 0-9
- ○ **D.** 3G is valid and equals 55 in denary

Q30 (3 marks)

Convert the denary number 237 to both binary and hexadecimal. Verify your conversions are equivalent.

- ○ **A.** Binary: 1110 1101, Hexadecimal: ED. Verification: E(14)x16 + D(13) = 224+13 = 237
- ○ **B.** Binary: 1110 1011, Hexadecimal: EB. Verification: E(14)x16 + B(11) = 224+11 = 235
- ○ **C.** Binary: 1111 0101, Hexadecimal: F5. Verification: F(15)x16 + 5 = 240+5 = 245
- ○ **D.** Binary: 1101 1101, Hexadecimal: DD. Verification: D(13)x16 + D(13) = 208+13 = 221

Total: **55** marks • Time: **30 min**

↑ Back to top

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